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DETAIL SPECIFICATION

Case Cartridge, 105MM, M14 and M217

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the manufacturing of the M14 and M217 case cartridge.

1.2 Classification. The cartridge case is described as follows:

1.2.1 Case cartridge M14.The M14 cartridge case can be used with 105MM cartridges using the M67 propelling charge.

1.2.2 Case cartridge M17. The M217 cartridge case can only be used with the cartridge, 105MM, HE, DPICM, M915/M916 using the M200 propelling charge.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following documents, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD preferred methods for acceptance of product

(Unless otherwise indicated, copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

7548409 - Case, Cartridge, 105MM,

12972264 - Case, Cartridge, Brass, 105MM, M217

7555278 - Gage, Chamber

(Copies of these documents may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM B 154 - Standard Test Method for Mercurous Nitrate Test for Copper and Copper Alloys
ASTM E 8 - Standard test Method for Tension Testing of Metallic Materials
ASTM E 18 - Standard test Method for Rockwell Hardness of Metallic Materials
ASTM E 340 - Standard test Method for Macroetching Metals and Alloys

(Copies of these documents are available online at http://www.astm.org or from American Society for Testing and materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Case Cartridge.

3.2.1 Case cartridge M14. M14 cartridge case shall comply with all the requirements specified on drawing Dwg.7548409

3.2.2 Case cartridge M217. M217 cartridge case shall comply with all requirements specified on Dwg.12972264

3.3 Ballistic functioning.

3.3.1 Cartridge Case M14. The M14 cartridge case shall function properly when assembled into a 105MM cartridge with an M28 series primer and M67 propelling charge and fired in an M101A1, M102 or M119A1 105MM Howitzer using standard field loading, firing and extraction procedures (remote firing permitted).

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3.3.2 Cartridge Case M217. The M217 cartridge case shall function properly when assembled into a M915 cartridge with an M28 series primer and M200 propelling charge and fired in an M119A1 105MM Howitzer using standard field loading, firing and extraction procedures (remote firing permitted).

3.3.3 Obturation. The cartridge case shall effectively seal the breech of the weapon against gases generated during firing. The cartridge case shall show no visual evidence of propellant gas migration beyond the forward most four inches of the exterior of the case after firing.

3.3.4 Soundness. The cartridge case shall exhibit the ability to withstand the maximum operating pressure without splitting, cracking or breaking up. Additionally, the cartridge case shall not exhibit any evidence of material loss or burn-through due to combustion or erosion due to propellant gases (see 6.8).

3.3.5 Extraction. After firing, the cartridge shall be extracted by the service extractor used during normal fielded operation.

3.3.6 Primer retention. The cartridge case shall retain its primer during firing and extraction. Additionally, there shall be no evidence of propellant blow-back after firing (see 6.8).

3.4 Chamber gaging. The cartridge case, after stress relieving, shall be compatible with the chambers of intended weapon systems as evidenced by successful testing with the chamber gage described in Dwg. 7555278. The maximum insertion force is to be a dead load of 15 pounds. No part of the base of the cartridge case shall protrude past the rear of the gage.

3.5 Mechanical properties.

3.5.1 Hardness. The metal at the location 0.120 inch minimum from the surface around the primer hole shall have a Rockwell hardness value of not less than B65.

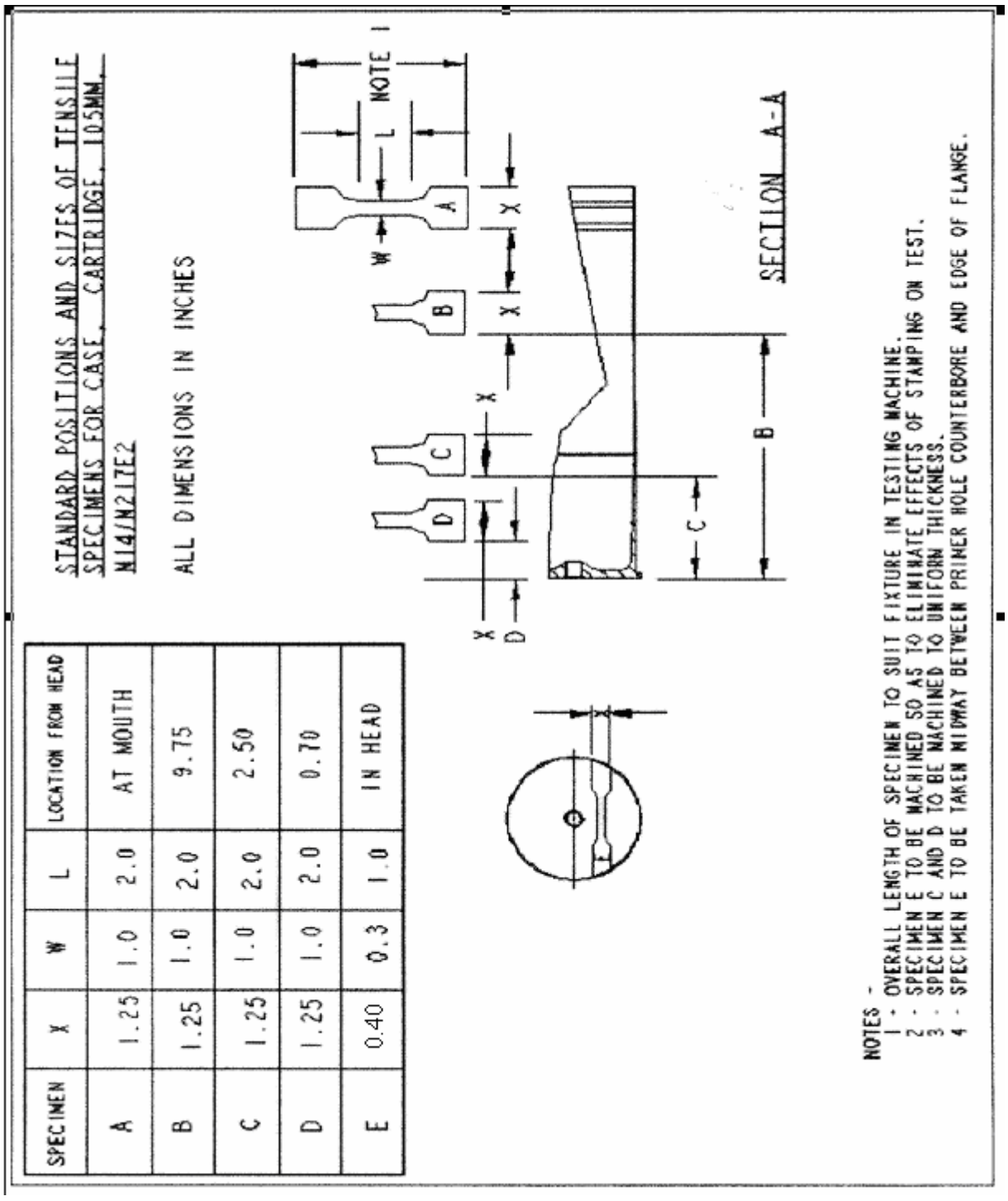
3.5.2 Tensile strength. Specimens taken from sample cartridge cases in accordance with Figure 1 shall comply with the requirements of Table 1 (see 6.10).

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TABLE I. Tensile strength test requirement

	Pounds per square inch (PSI)
Specimen A	44,000 – Minimum 58,000 Maximum
Specimen B	50,000 - Minimum
Specimen C	65,000 - Minimum
Specimen D	65,000 - Minimum
Specimen E	57,000 - Minimum

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SPECIMEN	X	W	L	LOCATION FROM HEAD
A	1.25	1.0	2.0	AT MOUTH
B	1.25	1.0	2.0	9.75
C	1.25	1.0	2.0	2.50
D	1.25	1.0	2.0	0.70
E	0.40	0.3	1.0	IN HEAD

Figure 1. Standard positions and sizes of tensile specimen for case, cartridge, 105MM, M14/M217E2

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3.6 Mercurous nitrate test. The cartridge case shall show no evidence of cracking within 24 hours after immersion in mercurous nitrate.

3.7 Metal defects. The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches, porosity, splits, inclusions, pipe, and other metal defects. The cartridge case shall be free from folds and cold shuts except as defined in 3.7.1. Visual standards may be utilized as required (see 6.5).

3.7.1 Macro examination. The cartridge case shall contain no folds and cold shuts in the radius area between the tangents from the base and sidewall (see figure 2). Folds and cold shuts in the tangent area shall not exceed .035 inches in length.

3.8 Workmanship. All parts shall be finished and assembled in a thorough, workmanlike manner. The parts shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the cleaning agent contaminate parts. All required markings shall be neat and sharply defined.

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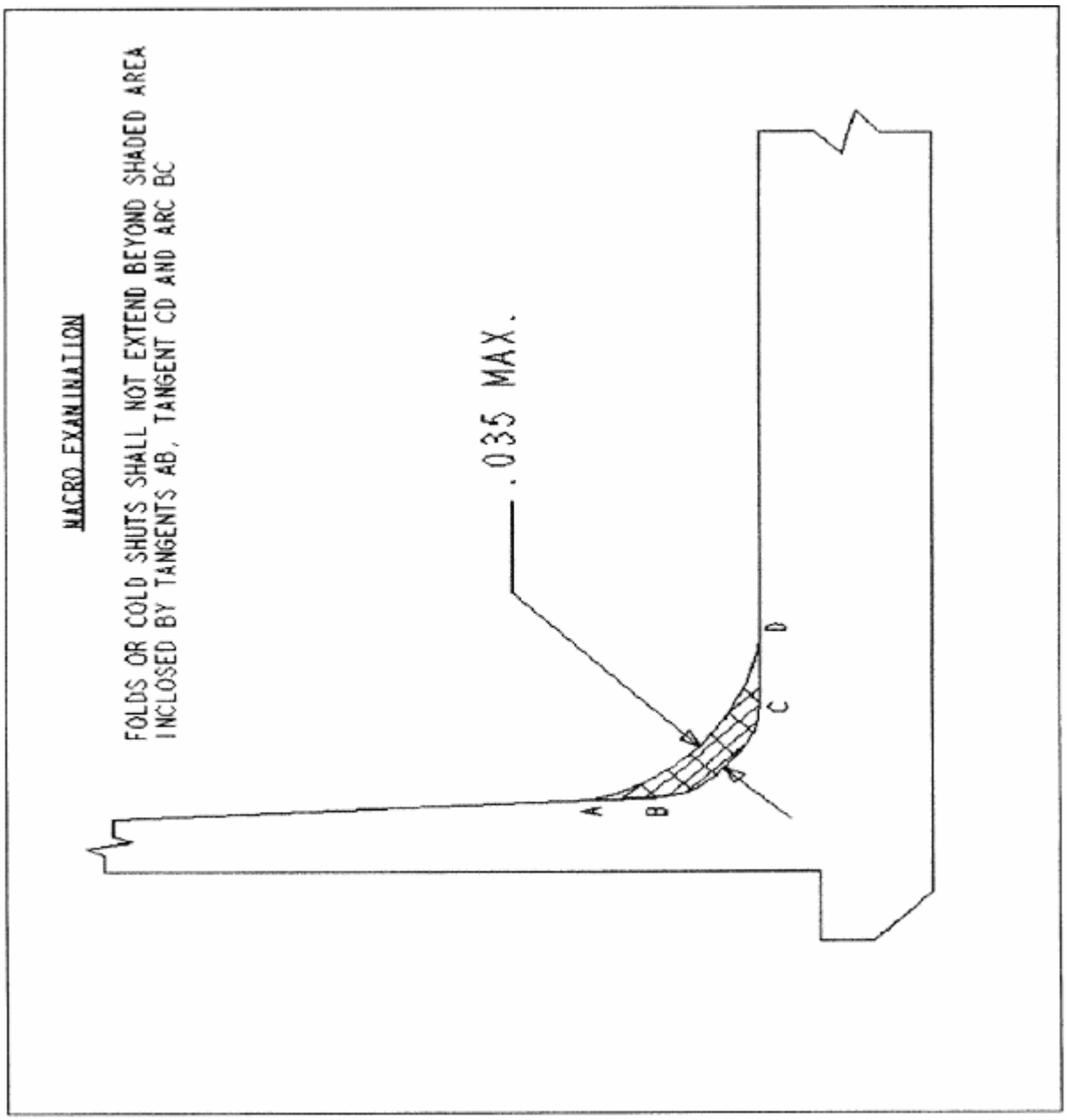


Figure 2. Macro examination

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4. VERIFICATION

TABLE II. Requirements/Verification cross reference matrix

METHOD OF VERIFICATION
N/A – NOT APPLICABLE
1 – ANALYSIS
2 – DEMONSTRATION
3 – EXAMINATION
4 - TEST

CLASSES OF VERIICATION
A – FIRST ARTICLE
B - CONFORMANCE

SECTION 3 REQUIREMENTS	PARAGRAPGH TITLE	VERIFICATION METHOD					VERIFICATION CLASS		SECTION 4 VERIFICATION
		N/A	1	2	3	4	A	B	
3.1	First Article				X	X	X		4.2 Table III
3.2	Case cartridge				X	X	X	X	4.3
3.2.1	Cartridge case M14				X	X	X	X	4.3.3.1/4.4
3.2.2	Cartridge case M217				X	X	X	X	4.3.3.1/4.4
3.3	Ballistic functioning					X	X	X	4.4.1
3.3.1	Cartridge case M14					X	X	X	4.4.1
3.2.2	Cartridge M217					X	X	X	4.4.1
3.3.3	Obturation					X	X	X	4.4.1a
3.3.4	Soundness					X	X	X	4.4.1b
3.3.5	Extraction					X	X	X	4.4.1c
3.3.6	Primer retention					X	X	X	4.4.1d
3.4	Chamber gaging					X	X	X	4.4.1e
3.5	Mechanical properties					X	X	X	4.4
3.5.1	Hardness					X	X	X	4.4.2
3.5.2	Tensile strength					X	X	X	4.4.3
3.6	Mercurous nitrate test					X	X	X	4.4.4
3.7	Metal defects				X		X	X	4.3.3.1
3.7.1	Macro examination				X		X	X	4.4.5
3.8	Workmanship				X		X	X	4.3.3.1

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4.1 Classification of inspections. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article. When specified, a sample of the cartridge cases shall be subjected to first article verification in accordance with Table III.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in Table III.

4.2.2 First article inspections to be performed. The first article verification shall be performed in accordance with Table III.

4.2.3 First article rejection. If any assembly, component or test specimen fails to comply with requirements, the first article sample shall be rejected.

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TABLE III. First article inspection

TITLE Case, Cartridge, 105MM, M14/Case, Cartridge, 105MM, M217	**SHEET** 1 **OF** 1		**DRAWING NUMBER**
	Number of FAT Samples	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Case, Cartridge (Dwg 9216352 or 13010426)			
Examination for defects Note 1,2 (Except Proving Ground Test)	20	3.2	4.3.3.1
Hardness	5	3.5.1	4.4.2
Tensile strength	5	3.5.2	4.4.3
Mercurous nitrate test	5	3.6	4.4.4
Macro examination	5	3.7.1	4.4.5
Proving Ground Test	30	3.3	4.4.1
Note 1. The examination for defects in paragraph 4.4.2 are for dimensional and visual inspections and in many cases must be witnessed during assembly. Note 2. Dimensions marked tool control shall be gaged during first article inspections			

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916 paragraph 4.2, Acceptance by Tables. Maximum lot size shall be 25,000 cartridge cases.

4.3.2 Heat treatment. Monitoring at unscheduled intervals shall be conducted for the processes of anneal, stress relief, and mouth anneal. Failure to comply with the applicable requirements shall be cause for rejection of all cartridge cases processed since the last monitored approval of the process.

4.3.3 Classification of characteristics

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, inspection level IV shall be used for all characteristics defined as majors and inspection level II for all minor characteristics. Critical characteristics shall be addresses in accordance with MIL-STS-1916.

b. Conformance Inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.3.1. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Tool control. Dimensions marked tool control shall be gaged after 250 items are produced from each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.3.1	TITLE Case, Cartridge	SHEET 1 OF 3	DRAWING NUMBER 7548409 or 12972264
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of primer hole with body basic diameter	Level IV	3.2	AIE
102	True position of primer counterbore with primer hole	Level IV	3.2	AIE
103	Diameter of flange, min	Level IV	3.2	AIE
104	Diameter of mouth	Level IV	3.2	AIE
105	Length from primer hole counterbore to forward surface of flange, min	Level IV	3.2	AIE
106	Thickness of wall at mouth	Level IV	3.2	AIE
107	Thickness of wall at basic length	Level IV	3.2	AIE
108	Thickness of flange, min	Level IV	3.2	AIE
109	Thickness of head, min	Level IV	3.2	AIE
110	Contour of inside of head and sidewalls not smooth	Level IV	3.2	Visual
111	Lot number not present or incorrect	Level IV	3.2	Visual
112	Macro examination	Note 1	3.7.1	4.4.5
113	Metal defective	Level IV	3.7	Visual (see 6.6)
114	Hardness	Note 1	3.5.1	4.4.2
115	Tensile strength	Note 1	3.5.2	4.4.3

NOTES:
1. 5 samples per lot. Accept on 0, reject on 1 defect.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.3.1	TITLE Case, Cartridge	SHEET 2 OF 3	DRAWING NUMBER 7548409 or 12977264
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
116	Maximum profile	100%	3.4	4.4.6
117	Excessive metal raised by lot number application	Level IV	3.2	AIE
118	Lot number illegible	Level IV	3.2	Visual
119	Diameter of primer hole	Level IV	3.2	AIE
120	Depth of primer hole counterbore	Level IV	3.2	AIE
121	Taper in primer hole	Level IV	3.2	AIE
122	Ballistic functioning	4.3.3.1	3.3	4.4.1
123	Mercurous nitrate test	Note 1	3.6	4.4.4

1. 5 samples per lot. Accept on 0, reject on 1 defect.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.3.1	TITLE Case, Cartridge	SHEET 3 OF 3	DRAWING NUMBER 7548409 or 12977264
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Diameter of primer hole counterbore	Level II	3.2	AIE
202	Diameter of forward obturating bead, min (M217)	Level II	3.2	AIE
203	Diameter of rear obturating bead, min (M217)	Level II	3.2	AIE
204	Length to forward obturating bead (M217)	Level II	3.2	AIE
205	Length to rear obturating bead (M217)	Level II	3.2	AIE
206	Length to basic diameter near mouth, min	Level II	3.2	AIE
207	Length to basic diameter near midbody, min	Level II	3.2	AIE
208	Length to basic diameter near head, min	Level II	3.2	AIE
209	Length total	Level II	3.2	AIE
210	Radii or chamfers missing or incorrect	Level II	3.2	Visual
211	Surface finish improper	Level II	3.2	Visual
212	Burr	Level II	3.8	Visual
213	Foreign matter	Level II	3.8	Visual
214	Evidence of poor workmanship	100%	3.8	Visual

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4.4 Methods of Inspection.

4.4.1 Ballistic testing. See Table IV for the verification test summary for this paragraph. The cartridge case test sample shall be assembled into a complete cartridge in the same manner as required for service firing. The propellant charge shall be such that it produces the maximum chamber pressure specified on the applicable drawing plus or minus 5 percent. The cartridge shall be temperature conditioned and fired in a serviceable weapon for which the round is standard. The weapon shall be operated in the service manner. Condition the cartridge to 70 +/-5 degrees F for a minimum of 12 hours. Assemble and fire the rounds, making sure the chamber pressure complies with the drawing requirements +/-5 percent. Inspect the cartridge cases for the following:

a. Obturation verification. Visually inspect for propellant gas migration as described in 3.3.3.

b. Soundness verification. Visually inspect for material failure in the fired cartridge case. Stretching and bulging is not considered a defect unless it results in the inability for the case to be extracted. Splits detected in the sidewall more than one caliber from the head are considered differently than the one detected within one caliber of the head.

c. Extraction verification. Conformance is measured by the ability of the cartridge case to be removed with the use of tools

d. Primer retention verification. Examination shall be made to verify the primer has not been forced out of the cartridge case to the extent that it is no longer held by cartridge case. Evidence that propellant gas migrated along the primer is also cause for rejection

e. Failure to chamber. If any cartridge fails to chamber for reasons other than handling with the cartridge which indicates the cartridge case is the improper size. The cartridge shall be set aside for examination, if it demonstrates a failure in the contractor’s inspection system.

4.4.1.1 Ballistic functioning sampling. Samples for ballistic functioning verification shall be taken as equally possible from each day’s production (see 6.9).

4.4.1.2 Lot acceptance ballistic performance sampling. Ballistic acceptance sampling shall be 30 cartridges. If test results indicate a retest, an additional 60 cartridges shall be required. See Table IV for acceptance criteria.

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TABLE IV. Ballistic verification matrix

	First Article Sample: 30	Accept	Reject				
A	Obturation	0	1				
B	Soundness (except as specified below)	0	1				
B	Sidewall cracks within one caliber of the head	0	1				
B	All other sidewall cracks	1	2				
C	Extraction	0	1				
D	Primer retention	0	1				
E	Failure to chamber	0	1				
	Lot acceptance test						
	Initial sample: 30	Accept	Retest	Reject	Retest sample:	Accept	Reject
A	Obturation	0	1	2	Additional	1	2
B	Soundness (except as specified below)	0	-	1	60 Score	N/A	
B	Sidewall cracks within one caliber of the head	0	-	1	Cumulative 90 unit result	N/A	
B	All other sidewall cracks	0	2	3		2	3
C	Extraction	0	1	2		1	2
D	Primer retention	0	1	2		1	2
E	Failure to chamber	0	-	1		N/A	

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4.4.2 Hardness. The cartridge case shall be sectioned through the center of the head. Three hardness readings shall be taken equally spaced through the head on both sides of the primer hole. The hardness readings shall be made in accordance with the procedures prescribed by ASTM E 18.

4.4.3 Tensile strength. The test specimens shall be taken from the cartridge case as indicated in Figure 1. The tensile strength shall be determined in accordance with the procedures prescribed by ASTM E 8.

4.4.4 Mercurous nitrate test. The mercurous nitrate test shall be performed as specified in ASTM B 154.

4.4.5 Macro examination. The cartridge case shall be sectioned through the center of the head, polished and examined for fold and cold shuts in the radius area if the head in accordance with the procedures prescribed by ASTM E 340.

4.4.6 Maximum profile and alignment gage. Each cartridge case after stress relieving shall freely enter a maximum profile and alignment gage, Dwg.7555278. The gate of the gage shall close without interference with the cartridge case head (flange).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M14 Cartridge Case covered by this specification is intended for use with certain 105MM Howitzer cartridges and M67 propelling charges to form projectiles. The M217 Cartridge Case covered by this specification is intended for use in the M915 and M916, 105MM howitzer cartridges only and the M200 propelling charge.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Packaging requirements (see 5).

c. Requirements for submission of first article sample (see 4.2).

d. Requirements for submittal of Acceptance Inspection Equipment (see 6.3)

e. Requirements for ammunition lot numbering and submission of ammunition data cards (see 6.4)

f. Requirements for approval of defect standards (see 6.5)

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards should be forwarded to Commander, ARDEC, ATTN: AMSRD-AAR-AEM-C and AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000.

6.5 Defect standards. Defect standards must be approved by the technical agency. All visual standards for defects will be established for each contract after the first month’s production, or whenever requested by contracting agency.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armaments, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.7 Process notes. The following notes are provided for information only:

a. Processing should assure that mechanical properties specified on the applicable drawings are uniform throughout each section.
b. Processing should ensure the entire cartridge case

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is thermally treated after the heading operation.

c. Processing should include adequate stress relief at the times and temperatures suitable for development of desired properties.

6.8 Definition.

a. Burn-through. Burn-through is an undesirable condition that is caused by the propellant gases eroding through the wall of the cartridge case. It is possible to see visual evidence of burn-through on the exterior wall of the cartridge case after firing, since the gases leave a smudge on that surface.

b. Blow-back. Blow-back is an undesirable condition that is caused by a poor seal at the primer interface. When the propelling charge is ignited, hot gases pass through a poor seal and leave burn marks on the exterior surface at the primer interface. These marks may also be visible on the breech of the weapon.

c. Obturation. Obturation is a desirable condition in a cartridge case. It is the expansion of the case sidewall to conform to the breech of the weapon in the area of projectile interface. The expanding propellant gases cause the expansion. Obturation provides necessary seal so that the projectile portion may reach the desired velocity to make range. Poor obturation is evidenced by gas erosion and smudging by the propellant gases on the exterior of the cartridge case after firing.

d. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard which is used to assure conformance of material to the contract requirements.

e. Heat treatment. Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and metallurgical properties of the metal involved.

6.9 Ballistic sampling. The requirement of the ballistic sample to represent each day’s production is to help verify process control throughout the lot. As a minimum at least one sample should be taken per day and the balance taken from the last day of production.

6.10 Tensile strength. It is not implied that cartridge cases that meet the minimum strength values will function satisfactorily in ballistic test

6.11 Superseding document. DTL 8838129 specification is a ARDEC unique specification that is not in ASSIST. If a copy is required it may be requested online at Drawing-request@pica.army.mil mailto:Drawing-request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000.

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6.12 Subject term (key word) listing.

Ammunition
Howitzer

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1315-2007-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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Check the source to verify that this is the current version before use.